CSBP711 ADVANCED ARTIFICIAL INTELLIGENCE · PAPER CRITIQUE + COURSE PROJECT · UAEU · FALL 2026

Does a Vision Transformer really beat CNNs on children's pneumonia X-rays?

A reality check of Singh et al., "Efficient pneumonia detection using Vision Transformers on chest X-rays", Scientific Reports 14:2487 (2024)

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Part 1 · Paper critique: we re-ran the paper's claim under its protocol and ours, on the same dataset and pipeline as our Assignment 1.
Part 2 · Project: can an LLM write a classifier's weights itself, with no training at all?

The question: two numbers on the same images, 11 points apart

Assignment 1 gave us a fair, patient-grouped result. The paper reports something much higher. One of two things is true.

OUR ASSIGNMENT 1

86 %

accuracy of the best model (ResNet-50), AUROC 0.971. Patient-grouped split, duplicates removed, the dataset's own test folder, two seeds.

SINGH ET AL. 2024

97.6 %

accuracy of a Vision Transformer (DeiT-Base), AUC 0.966. Random 80/10/10 split, one run, fifteen CNNs reported at 76–94 %.

Either the transformer is genuinely better, or the number comes from how it was evaluated.

Our pipeline can separate the two: run their model under our protocol, and their protocol on our models.

The paper in one line: problem, data, model, claim

Singh, Kumar, Kumar, Verma, Abhishek & Selvarajan. Scientific Reports 14:2487, January 2024. No code released.

1 → 2 → 3 → 4 → 5

Problem

Is this child's frontal chest X-ray NORMAL or PNEUMONIA?

Data

Kermany et al. 2018: 5,856 images, Guangzhou Women and Children's Medical Center, 73 % pneumonia

Split

Random 80 / 10 / 10 by image: 4,684 train, 586 val, 586 test. Patients not mentioned.

Model

DeiT-Base/16 at 224 px, ImageNet weights, fully fine-tuned: Adam, LR 1e-5 × 0.995/epoch, batch 16, 30 epochs

Claim

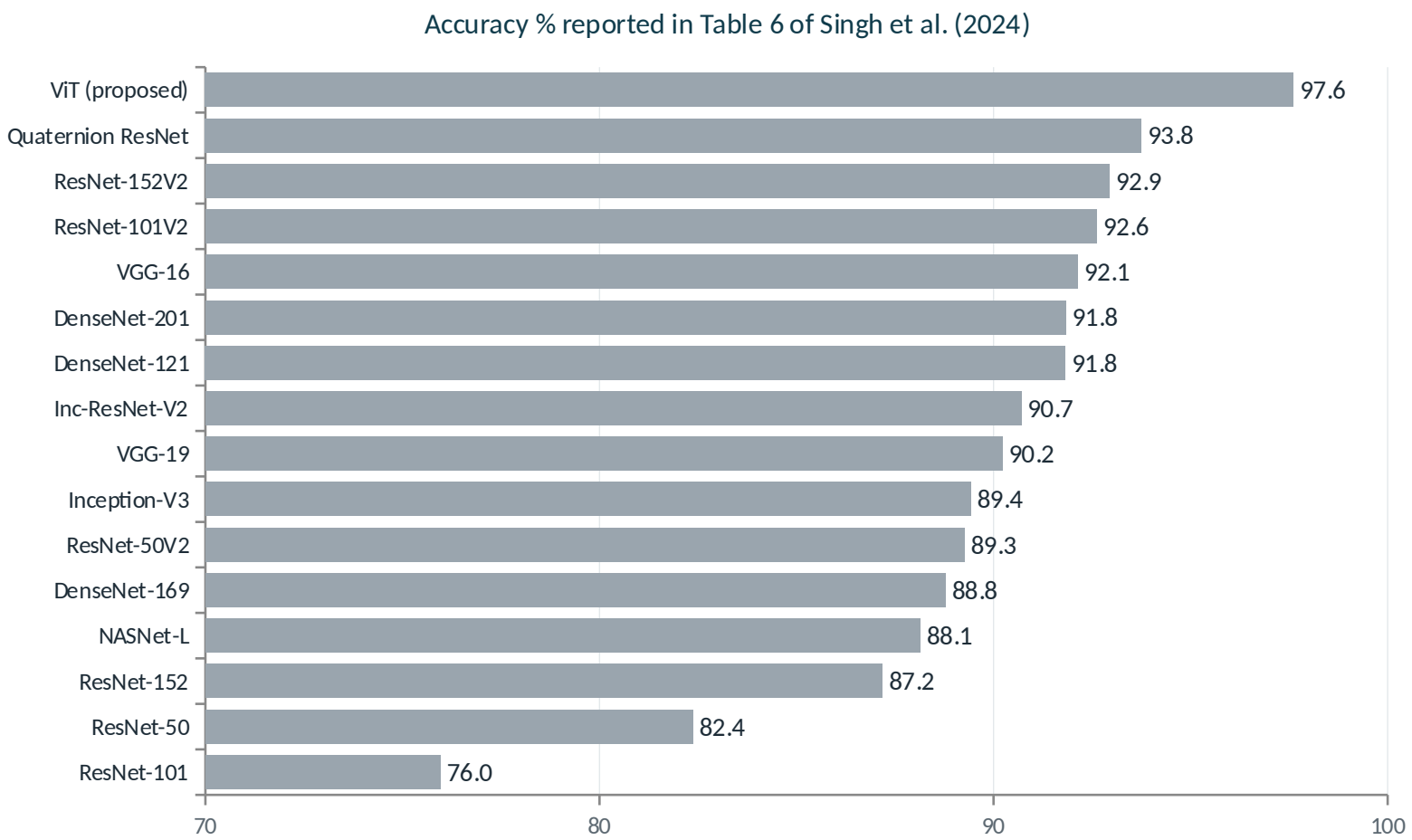
97.61 % accuracy, sens. 0.949, spec. 0.981, AUC 0.966: "a great improvement over all other architectures"

What they offer as evidence

One training run on one split. Accuracy with a Wald 95 % CI, a confusion matrix, ROC and precision–recall curves, and Table 6: fifteen CNN architectures scoring 76–94 % under what the text calls the same hyperparameters.

Their evidence: Table 6, fifteen CNNs against one transformer

Accuracy as reported by Singh et al. The ViT is 4.7 points above the best CNN and 15 points above ResNet-50.



What the table says

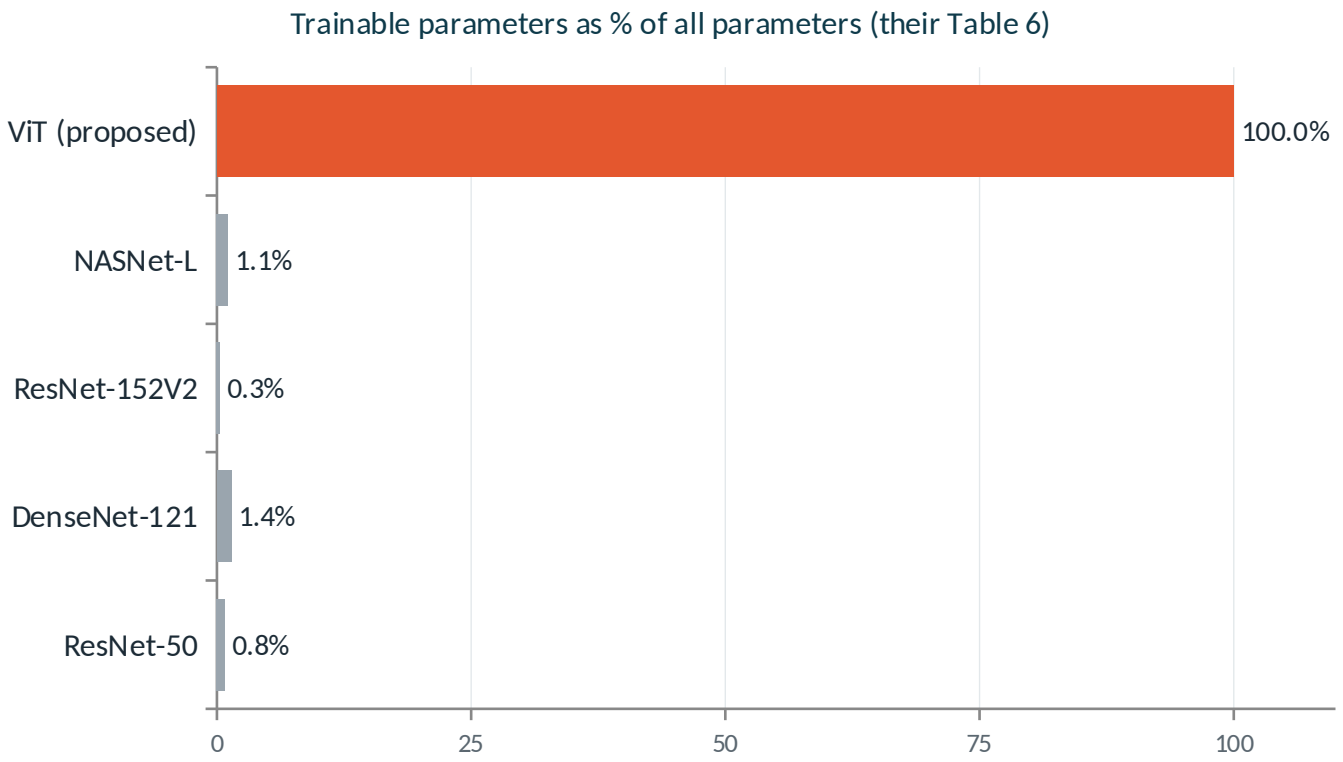
The transformer wins by a wide margin over every CNN, including architectures far larger than it (NASNet-L, 85 M).

What the table also says

A column most readers skip: trainable parameters. ResNet-50: 200,706. DenseNet-121: 100,354. ViT: 85,800,194. Next slide.

Weak point 1: CNN baselines frozen, transformer fully trained

Share of each model's parameters that were trained, from the two parameter columns of their Table 6.



0.8 %

of ResNet-50 was trained: a 200 k head on a frozen 23.6 M backbone

100 %

of the ViT was trained: all 85.8 M parameters

So the comparison is

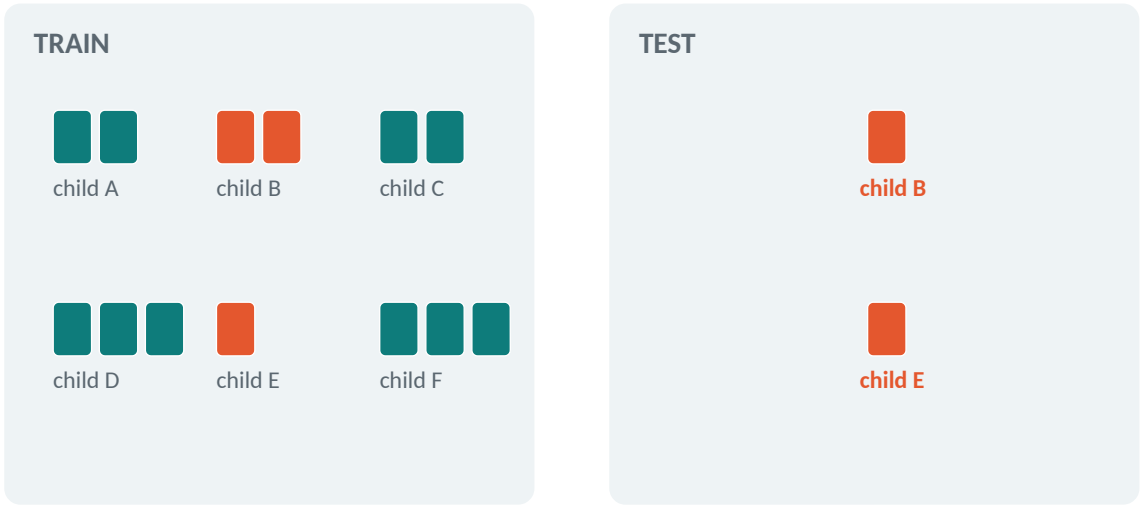
fine-tuning vs feature extraction, not transformer vs CNN. "All hyperparameters were kept the same" is true of the optimiser, not of the training.

Also: a 'Ref.' column sits beside every baseline, so some CNN numbers may be cited rather than run. The paper does not say.

Weak point 2: a random split by image puts one child on both sides

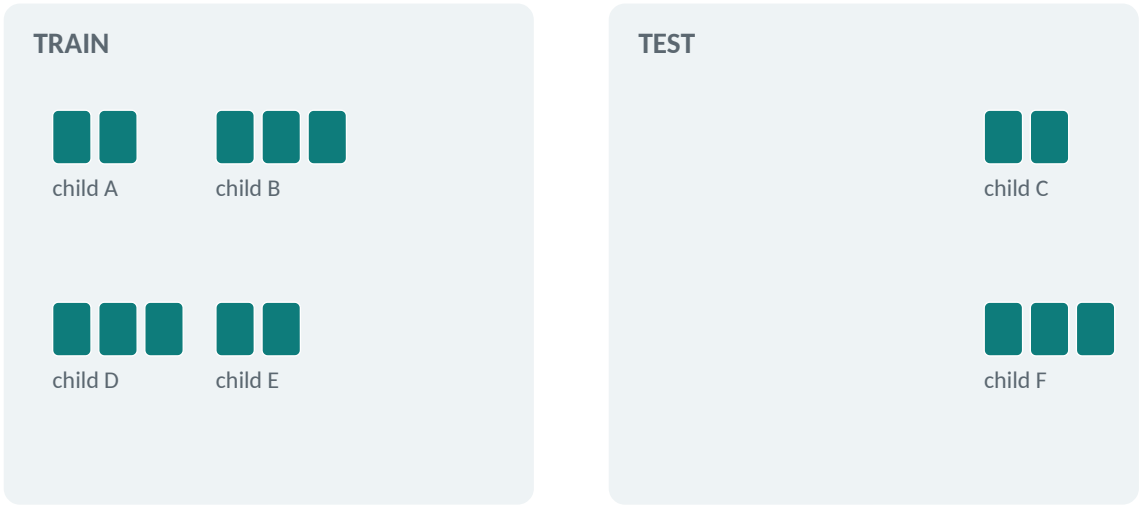
1,155 children in this dataset have more than one X-ray, up to 30. The paper never mentions patients.

Their split: random by image



A scan of child B and child E is in the test set while their other scans are in training. The model can recognise the child instead of the disease.

Our split: whole children



Every scan of a child is on one side only. The model must generalise to children it has never seen.

58 %

of test images have a sibling scan in training when we reproduce their split (336 of 583)

0 %

under our patient-grouped split

And their test set comes from the training pool

The dataset's own test folder is a different collection: in A1 every model scored ≥ 0.99 AUROC on validation but 0.90–0.97 on it. A random 10 % never faces that shift.

Weak point 3: the numbers do not reconcile, and the evidence is thin

Small inconsistencies are not fatal on their own. Together they show the evidence was not checked carefully.

Table 3 says

Test set: 330 pneumonia, 256 normal (586 images).

≠

The confusion matrix says

TP 152 + FN 8 = 160 pneumonia; TN 420 + FP 6 = 426 normal.

Same test set, different composition.

Also

F-score 0.952 (Table 5) vs 0.950 (Table 6) · cross-entropy vs binary cross-entropy · CI 96.2–98.9 % stated, 96.4–98.9 % by their own Eq. 7 · 5,863 images in text, 5,856 in tables

One run

No seed, no cross-validation, no variance. Test accuracy is tracked every epoch next to validation ("improves along with test accuracy"), which risks selecting on the test set.

Accuracy as headline

On a set that is 56–73 % pneumonia, a model that always says 'pneumonia' already scores the majority share. AUROC or balanced accuracy would be the honest lead metric.

Limitations and reproducibility

Limitations: one sentence ("data scarcity and the need for real-world validation"). No code, no split files, no seeds. The experiment can only be approximated.

WHAT THE EVIDENCE SUPPORTS, BEFORE WE RUN ANYTHING

The claim is larger than the numbers

Supported

"A fully fine-tuned DeiT-Base reaches about 98 % accuracy on a random image-level split of this dataset."

Not supported

"A Vision Transformer outperforms CNN architectures on pediatric pneumonia detection."

Strengths, for balance: a clear and falsifiable claim, a public dataset, a complete hyper-parameter table, a confusion matrix we can recompute from, and a cost note. That is what makes the claim testable.

Reality check design: two models × two protocols, one pipeline

Everything from Assignment 1 is unchanged. We added their model, and a switch that reproduces their split.

	A1 protocol patient-grouped split, official test folder (n = 618)	Paper protocol random 80/10/10 split by image (n = 583)
ViT DeiT-B/16 (their model)	**NEW · fair test of their model** Does the ViT beat ResNet-50 when both are fine-tuned under one protocol?	**NEW · reproduce their number** Do we get their 97.6 % with their model and their split?
ResNet-50 (our A1 winner)	**A1 result** 0.971 AUROC, 86.2 % accuracy. Our reference point.	**NEW · their split on our model** Does a fine-tuned ResNet-50 also reach ~97 % under their split?

Gaps we could not close: their 30 epochs (we keep 12), plain resize (we pad), Adam (we use AdamW). All three can only help their number; the report lists them.

Their model, faithfully

timm deit_base_patch16_224, ImageNet weights, fully fine-tuned, LR 1e-5 as in the paper. 85.8 M params.

Their split, faithfully

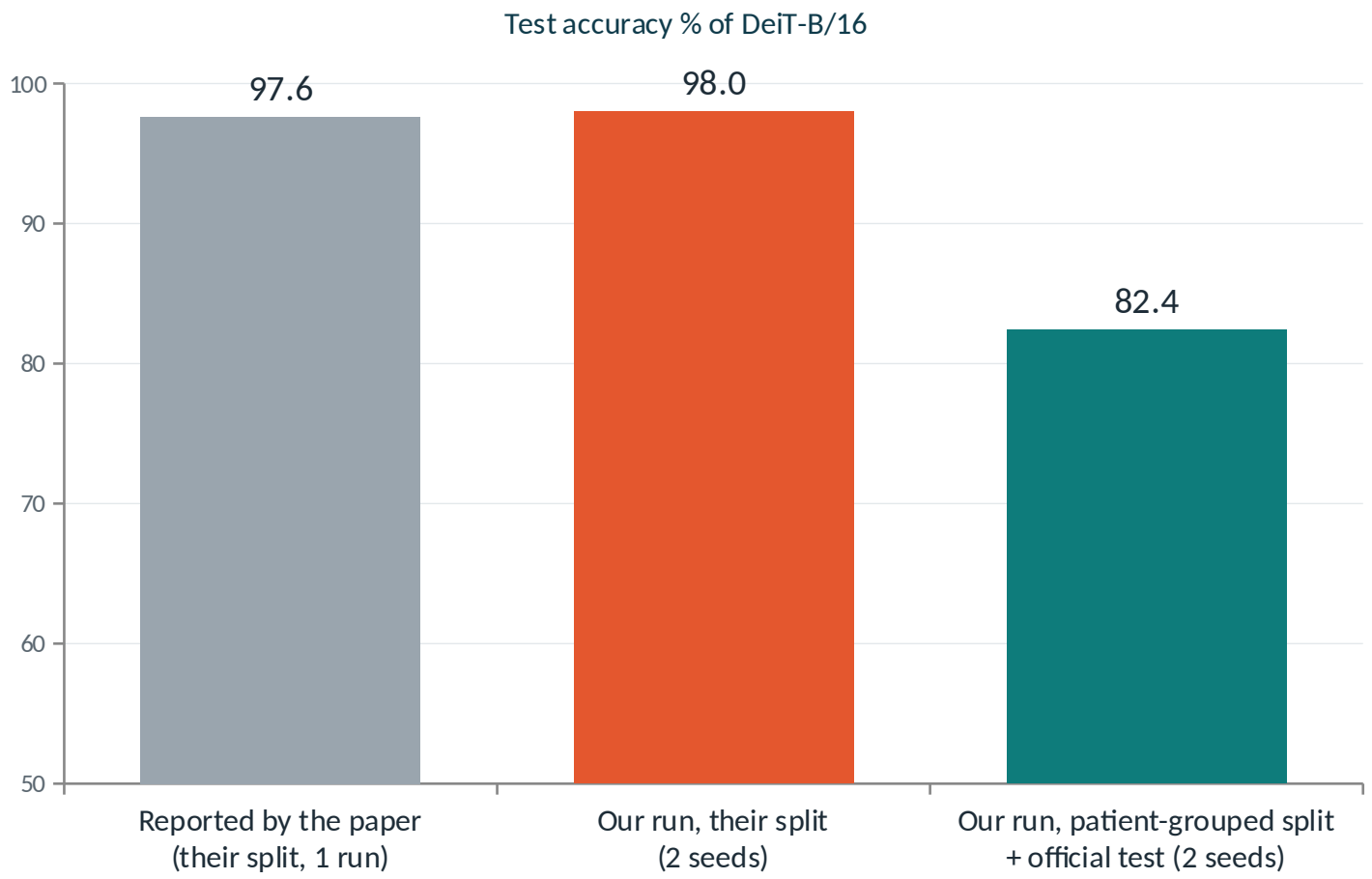
Pool the 5,824 kept images, random label-stratified 80/10/10 by image: 4,659 / 582 / 583. Leakage measured: 58 %.

Held fixed

Grayscale, pad-to-square, 224 px, same augmentation, AdamW, 12 epochs, best-val-AUROC checkpoint, seeds 42 + 43, one T4.

Result 1: their 97.6 % reproduces, and it depends entirely on the split

The same ViT, the same pipeline. Only the split changes.



98.0 %

our ViT under their split; they report 97.6 %. Specificity 0.97 vs their 0.98. Reproduced.

82.4 %

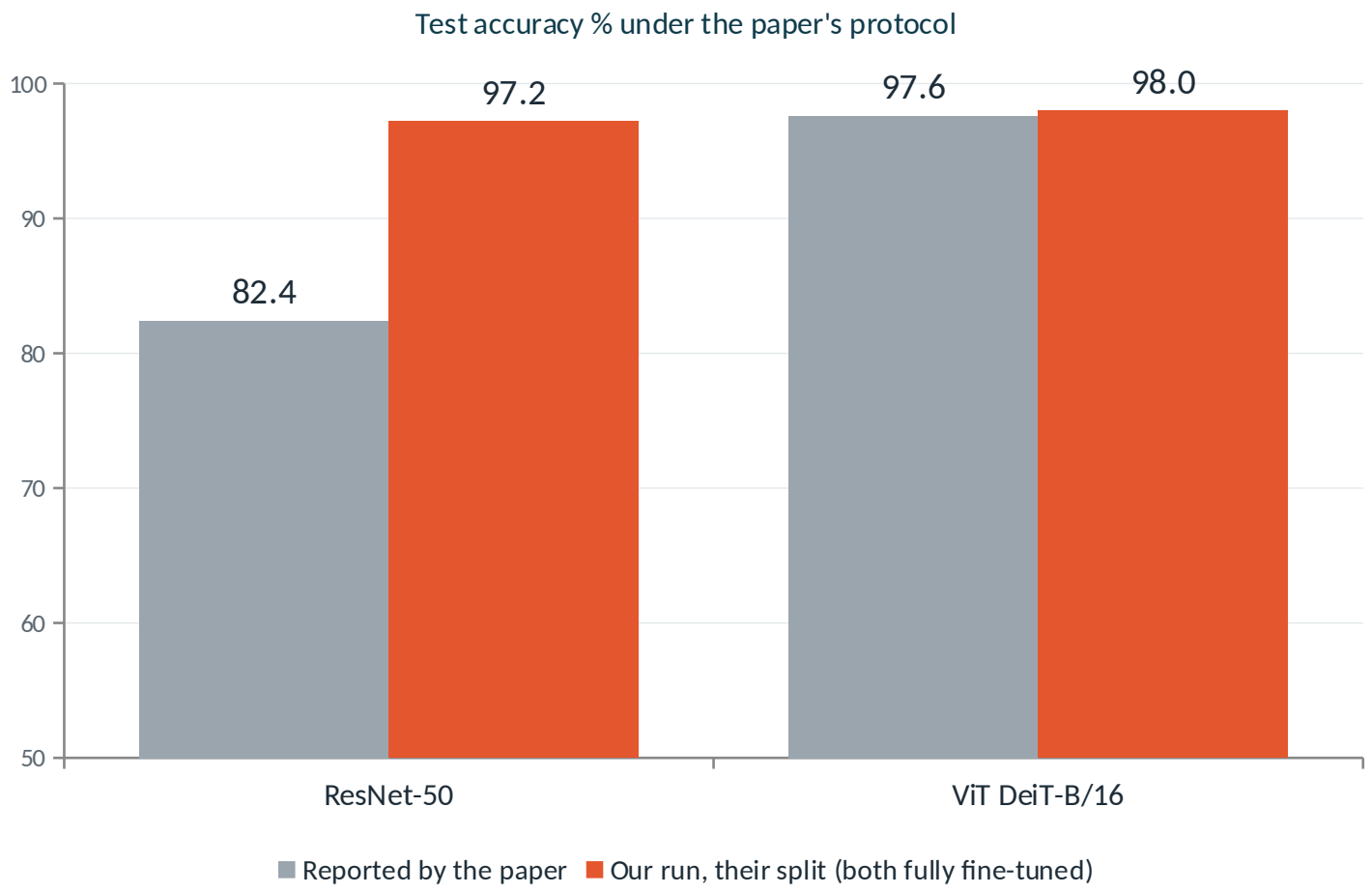
the same ViT under the fair protocol. Specificity 0.53.

Read it as

The number is real, and we got it. But it needs a test set that shares children with training and is drawn from the same collection. Remove both, and 15.6 points disappear. The model did not change; the evaluation did.

Result 2: under their split, a plain ResNet-50 gets the same number

The 15-point gap in their Table 6 is the frozen backbone, not the architecture.



97.2 %

fine-tuned ResNet-50 under their split (two seeds)

+14.8

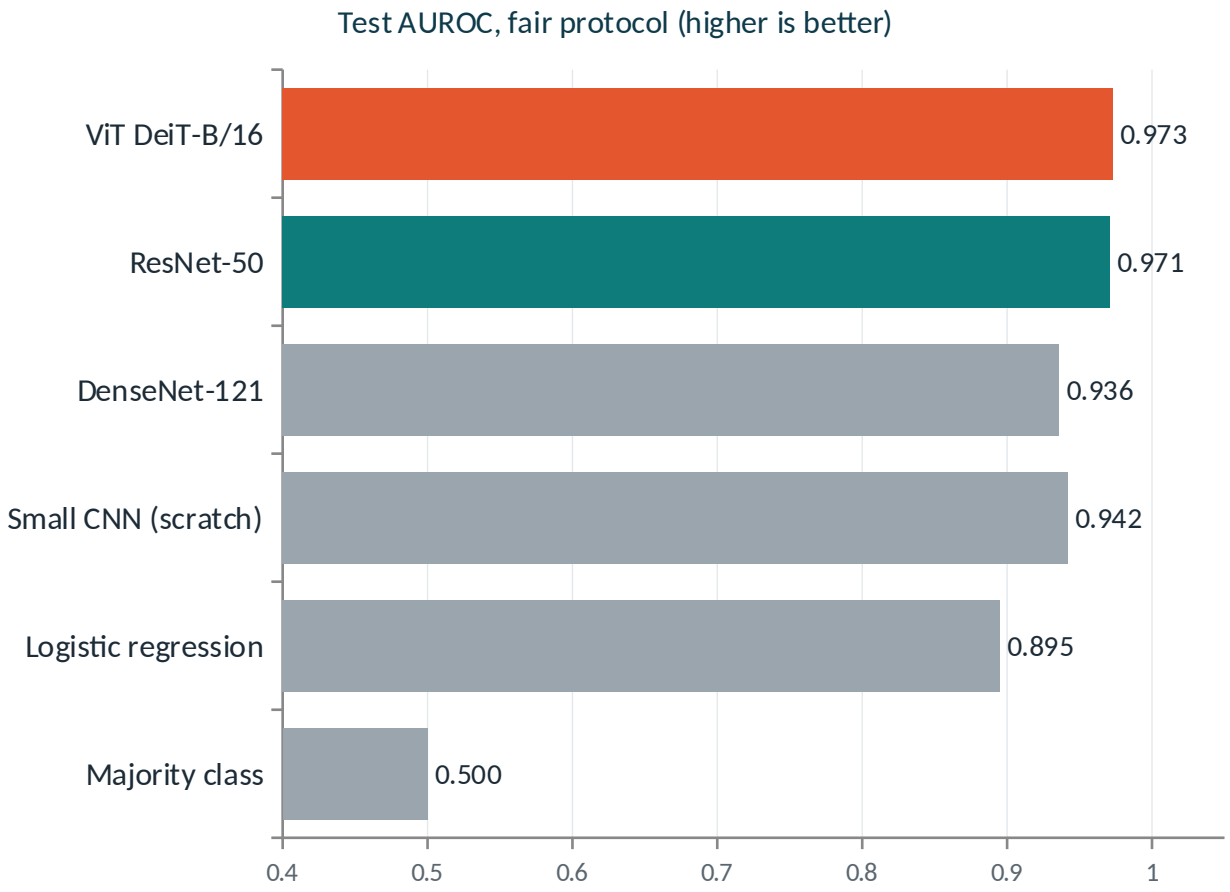
points above the 82.4 % they report for ResNet-50 with a frozen backbone

Read it as

Give the CNN the same treatment as the transformer and the advantage is 0.8 points, inside the seed-to-seed spread. Their comparison measured whether the backbone was trained, and called it architecture.

Result 3: fair protocol, the ViT ties ResNet-50 at 3.7× the size

All six models, patient-grouped split, official test folder, mean of seeds 42 and 43. Our Assignment 1 table with the paper's model added.



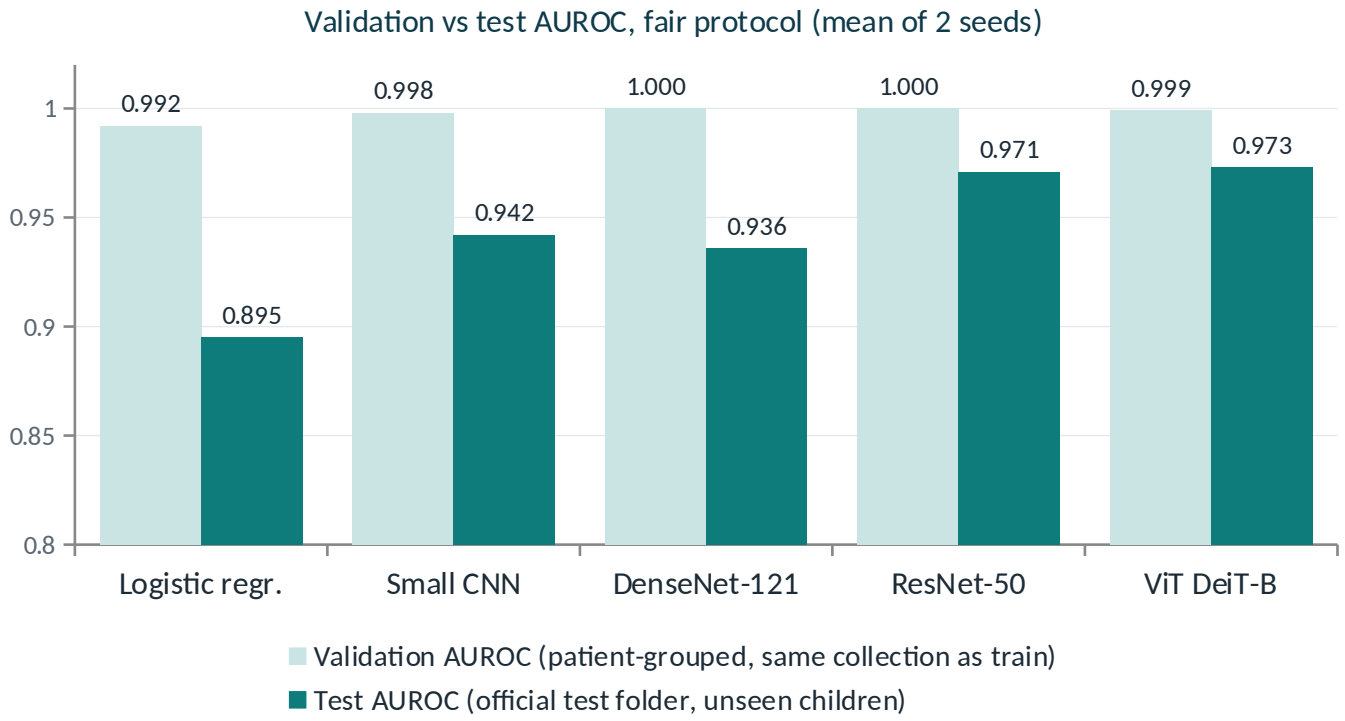
Model	AUROC	Acc. %	Spec.	Params	Train
ViT DeiT-B/16	**0.973 ± .001**	82.4	0.53	85.8 M	9.2 min
ResNet-50	**0.971 ± .001**	86.2	0.64	23.5 M	4.4 min
Small CNN	0.942 ± .006	82.6	0.54	0.39 M	3.4 min
DenseNet-121	0.936 ± .017	82.2	0.53	7.0 M	5.9 min
Logistic regression	0.895 ± .006	77.4	0.43	0.15 M	3.0 min
Majority class	0.500	62.6	0.00	0	0

A tie, at a price

AUROC 0.973 vs 0.971: the seed-to-seed spread is 0.001, so this is a tie. The transformer uses 3.7× the parameters and 2× the training time, and is worse on accuracy and specificity. Sensitivity is ≥ 0.98 for every model; the differences are all in the healthy children.

Why: two properties of the data explain the whole gap

Every model scores ≥ 0.99 AUROC on validation. The drop happens only when the test set is the official folder with unseen children.



Property 1 · Patient leakage

Under their split 58 % of test images have a sibling scan in training. Specificity jumps from 0.53–0.64 to 0.97–0.98 the moment that is allowed. Memorising a child is rewarded as accuracy.

Property 2 · Collection shift

The official test folder is a different collection (A1 audit: different image geometry and class prior). A random 10 % of the training pool never meets it, so their test is easier by construction.

Plus one property of their setup: frozen baselines.

Trained equally, ResNet-50 and ViT are within 0.8 points under their protocol and 0.002 AUROC under ours. Architecture did not move the number; the protocol did.

VERDICT

The claim does not hold up. The protocol, not the architecture, produced the number.

Reproduced

Their 97.6 % → our 98.0 % under their split. Their evidence is real for the question they asked.

Explained

A fine-tuned ResNet-50 gets 97.2 % under the same split. Patient leakage (58 %) and a same-collection test set inflate every model alike.

What survives

"On this dataset the evaluation protocol dominates the architecture." Under a fair protocol the ViT ties ResNet-50 at 0.97 AUROC, at 3.7× the size.

PART 2 · COURSE PROJECT

Can an LLM skip training altogether, and write the weights itself?

Part 1 showed that the protocol decides what a number means. Part 2 keeps our fair protocol fixed and changes something more radical: who sets the weights.

The inspiration

Takehi, Teoh, Zhang & Khattab (MIT CSAIL, Nov 2026), "Sparks of Weight-Space Programming": an LLM read and wrote neural-network weights directly. Its hand-written game policies beat networks trained with PPO for 100 M steps. (Blog post, not yet peer reviewed.)

Our question

On our pneumonia task, can a frontier LLM (Astra) write the weights of a small CNN, with no training at all, and match the same CNN trained by gradient descent?

What the MIT work did, and which part we can copy

Three modes. We copy the 'write from scratch' mode exactly: their Exp. 5 (one reply) and Exp. 6 (a budget of practice forward passes).

Mode	What the LLM got	Largest network	Their result	Our version
Read	Raw weights; "what was this trained on?"	7.2 M	Named game, shape or language	Not used
Write from scratch	Game rules, game source code, network shape; no data	1.5 M	MinAtar policies beat PPO trained for 100 M steps	**Conditions A and B**
Edit	A failing trained network + a few forward passes	0.8 B	Qwen MATH-500 7.9 → 48.1 %	Not used

Why their games worked

The rules and the simulator's source code fully define what a good move is. The LLM effectively compiled a strategy into weights, the way a programmer writes code.

What is different for pneumonia

There is no source code for 'pneumonia'. What separates the classes is defined only by labelled examples, which is exactly what training uses. So this is a hard test of the idea, and an informative one either way.

Setup: Astra writes all 5,569 weights of one small CNN

The same network is also trained normally, so the comparison is exactly the same 5,569 slots filled two ways.



A · Zero-shot (their Exp. 5)

One reply. Astra gets the task description, the preprocessing steps and the network code. No images, no data, no feedback, no tools.

B · Practice passes (their Exp. 6)

Same prompt, plus up to 20 practice passes: Astra sends a weights file, we run it once on 256 validation images and return the scores and labels. It never sees an image. It used 10 passes.

Rules that keep it honest

No gradients, no fitting, no automated search: every change must be designed and explained. Only numpy and math, no random numbers (checked automatically). Dataset not named; no image sizes shown. Test set scored once, after both conditions.

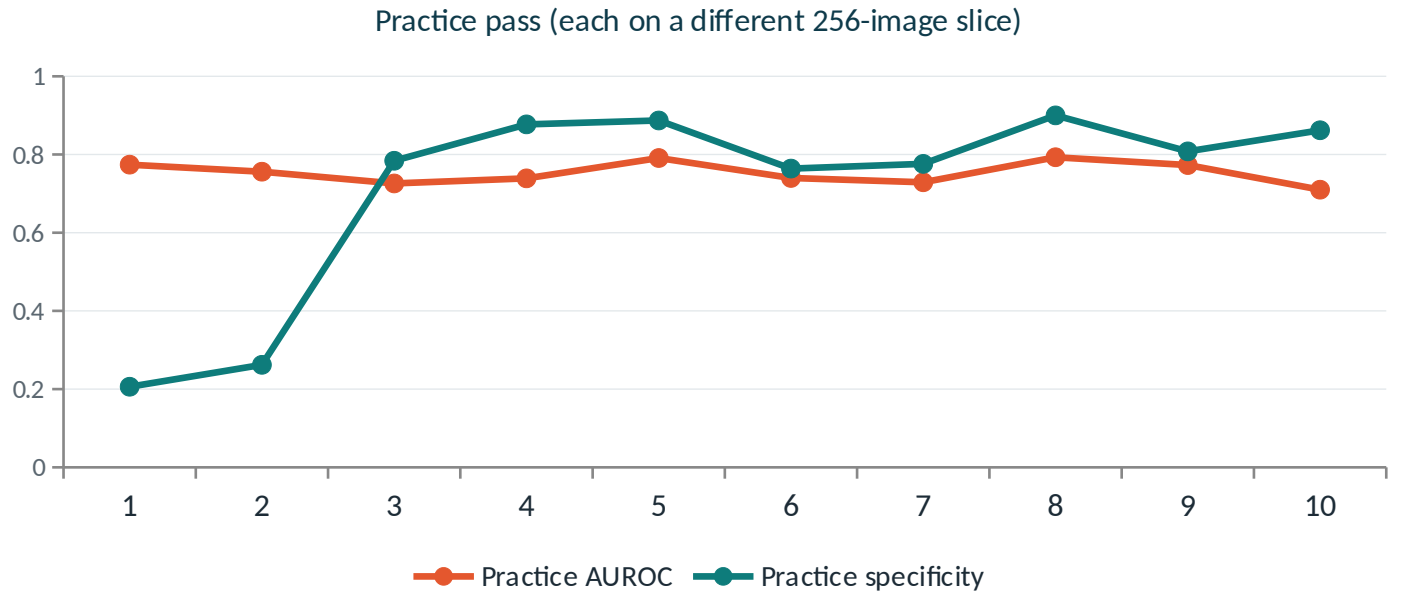
What Astra wrote, and how it changed it over 10 practice passes

A hand-designed detector: dark, air-filled lung is evidence against pneumonia; black padding and the heart are ignored.

Final design: weight of each 4 × 4 grid cell

0	0	0	0
0	1	1	0
0	1	0.5	0
0	0	0	0

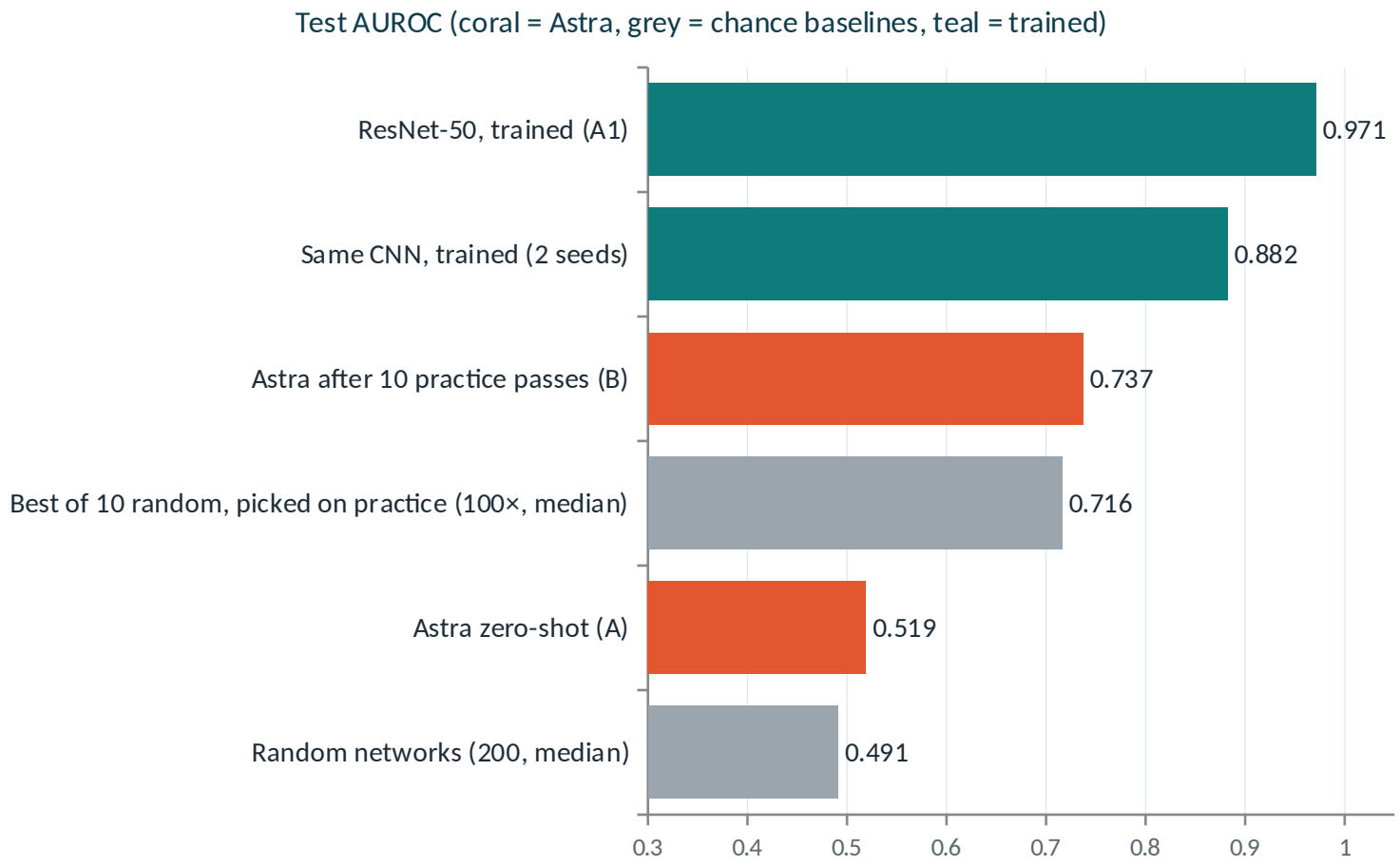
Lungs (rows 1–2) count; the lower-right cell is halved for the heart; borders and padding are ignored. Seven brightness thresholds, four 'dark region' indicators and one 'not black padding' indicator: 20 of 32 conv2 channels unused.



Pass	Change Astra tested	Kept?
3	Remove the prior bias (log 3 → 0): fewer false alarms	Yes
4–5	Brightness reference from the central chest, to cancel exposure differences	Yes · FINAL
6–7	Average dark-region evidence over neighbours	No
8–10	Narrower vertical smoothing, to keep gaps between ribs	Tie → no

Result: zero-shot is chance; with feedback, Astra ≈ random search

Test AUROC on the same 618 official test images, patient-grouped. No training in any bar except the last two.



0.519

Astra zero-shot, 95 % CI 0.47–0.57: no better than chance (48 % of random networks do as well)

0.737

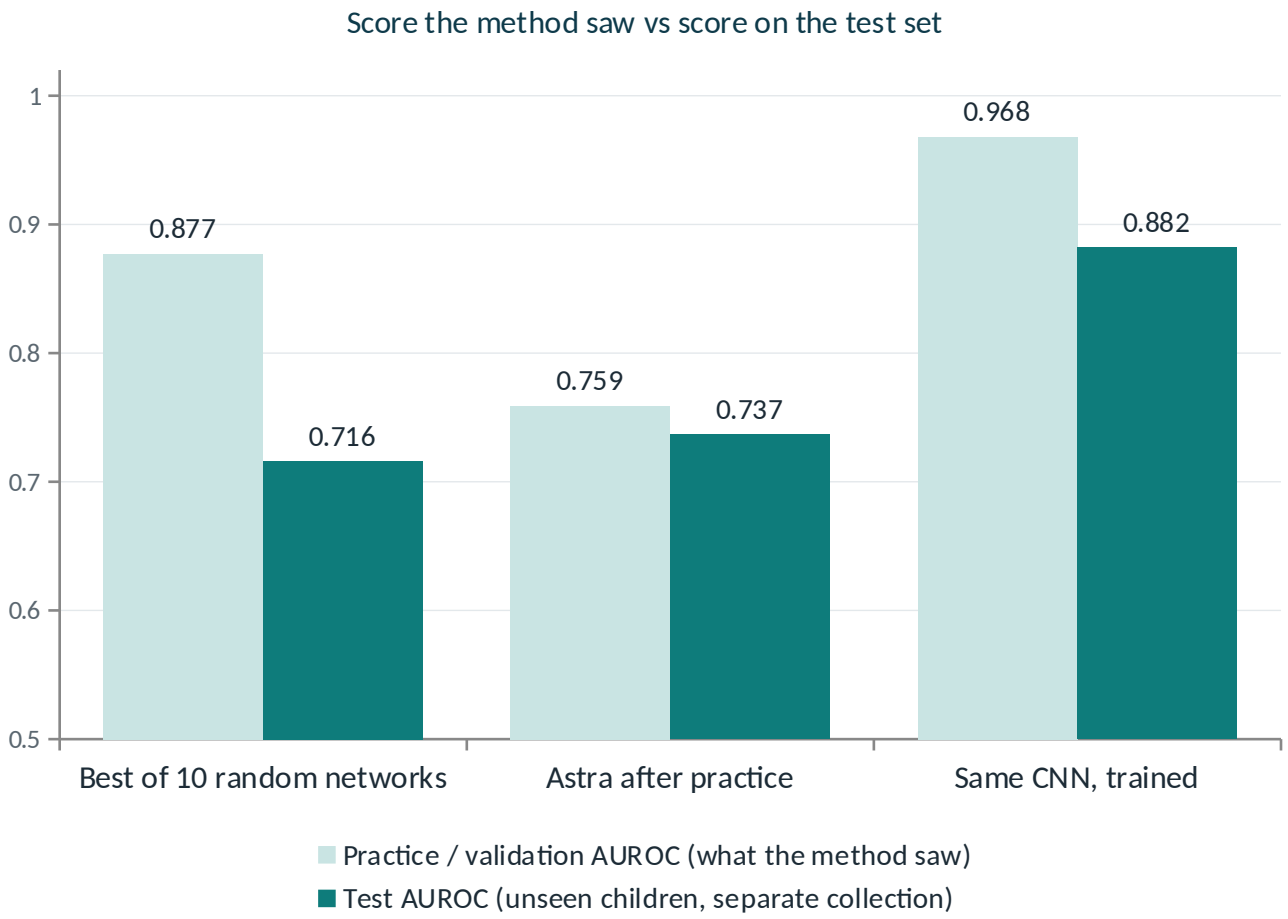
Astra with feedback, 95 % CI 0.70–0.78. It predicted 0.75 for itself.

Read it as

Feedback did the work, not the reasoning: picking the best of 10 random networks with the same feedback reaches 0.716, and beats Astra in 28 % of repeats. Training the same 5,569 weights reaches 0.882 in 3 minutes.

Why: rules can be written down; a diagnosis must be learned

And one thing Astra did better than random search: its practice score was honest.



1 · Knowledge is not a specification

MIT's games came with rules and source code that define a good answer. 'Dark lung = healthy' is true, but too coarse to separate these X-rays; the fine pattern lives only in labelled data.

2 · Random search was lucky; Astra was not

Random winners looked great on practice (0.88) and fell to 0.72 on test: picked by chance. Astra's design dropped only 0.02, and it forecast its own score.

3 · Training still wins clearly

Same 5,569 weights, 0.15 AUROC higher, 3 minutes on a T4. The MIT result does not carry over to a task defined by examples rather than rules.

CONCLUSION · BOTH PARTS

In both experiments, the evaluation decided what the number meant

Part 1 · Paper critique

The ViT's 97.6 % reproduces under a random image split, and a fine-tuned ResNet-50 gets 97.2 % under the same split. With patient grouping and the official test folder both fall to 82–86 %, and tie at 0.97 AUROC. The protocol, not the architecture, produced the claim.

Part 2 · Project

An LLM wrote a working classifier with no training (0.737 AUROC), but only with feedback, and it was no better than random search given the same feedback. Without the chance baselines, 0.737 would have looked like a breakthrough. Training still wins: 0.882.

What we would do next

For both parts: cheap runs that close the remaining gaps first, then the experiments that could change the answer.

1 · Close the last gaps

One run each under the patient split: their 30-epoch budget, plain resize instead of padding, a third seed. Expectation: ranking unchanged.

Run the image split on the three remaining A1 models to show the inflation is model-independent.

2 · Test where a ViT could win

An external pediatric test set (VinDr-PCXR) instead of this dataset's own folder, and a training-size sweep. ViTs are known to need more data than CNNs; 4,400 images is small, and that may be the real story.

3 · Weight-writing, next

Try the MIT 'edit' mode instead: let the LLM read our trained ResNet-50 and fix its false alarms (specificity 0.64). Repeat A and B with more practice passes and a second LLM, against the same random-search baseline.

Our question for another group

"In the paper you critiqued, were the baselines trained with the same budget and the same number of trainable parameters as the proposed model, and how would you check that from the paper alone?"

Backup: every number in one table

Mean ± std over seeds 42 and 43. n/r = not reported by the paper. Result files: results/*.json in the repository.

Model	Protocol	AUROC	Accuracy %	Sens.	Spec.	Params	Train
ViT DeiT-B/16 (reported)	Singh et al., random by image, n=586, 1 run	0.966	97.6	0.949	0.981	85.8 M	n/r
ViT DeiT-B/16 (ours)	Singh et al. protocol in our pipeline, n=583	0.997 ± .002	98.0 ± 0.6	0.984	0.972	85.8 M	9.0 min
ResNet-50 (ours)	Singh et al. protocol in our pipeline, n=583	0.995 ± .004	97.2 ± 0.4	0.969	0.978	23.5 M	5.0 min
ResNet-50 (reported)	Singh et al., backbone frozen	n/r	82.4	n/r	n/r	0.2 M trainable	n/r
DenseNet-121 (reported)	Singh et al., backbone frozen	n/r	91.8	n/r	n/r	0.1 M trainable	n/r
ViT DeiT-B/16 (ours)	A1: patient-grouped, official test, n=618	0.973 ± .001	82.4 ± 2.7	0.999	0.530	85.8 M	9.2 min
ResNet-50 (A1)	A1 protocol	0.971 ± .001	86.2 ± 3.4	0.997	0.636	23.5 M	4.4 min
DenseNet-121 (A1)	A1 protocol	0.936 ± .017	82.2 ± 2.7	0.999	0.526	7.0 M	5.9 min
Small CNN (A1)	A1 protocol	0.942 ± .006	82.6 ± 0.3	0.996	0.541	0.39 M	3.4 min
Logistic regression (A1)	A1 protocol	0.895 ± .006	77.4 ± 4.2	0.982	0.426	0.15 M	3.0 min
Majority class (A1)	A1 protocol	0.500	62.6	1.000	0.000	0	0